

Candidate Name	CT Group	Candidate Number



VICTORIA JUNIOR COLLEGE

JC 2 PRELIMINARY EXAMINATION 2008

H1 CHEMISTRY

8872/2

Paper 2

Monday
08/09/2008

2 hours
80 marks

Additional materials:

Answer Paper
Graph Paper
Data Booklet

INSTRUCTIONS TO CANDIDATES

Write your *name*, *CT group* and *candidate number* in the spaces at the top of this page.

Answer **all** questions in Section **A**. Write your answers in the spaces provided on this question paper.

Answer any **two** questions in Section **B**. Begin each question on a fresh sheet of paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are advised to show all working in calculations.

You are reminded of the need for good English and clear presentation in your answers.

FOR EXAMINER'S USE			
Section A		Section B	
1		1	
2		2	
3		3	
4		Total	/ 80

This question paper consists of **14** printed pages.

[Turn over

Section A

Answer **all** the questions in this section in the spaces provided.

1(a)

(i) Define the term *relative atomic mass*.

.....

.....

.....

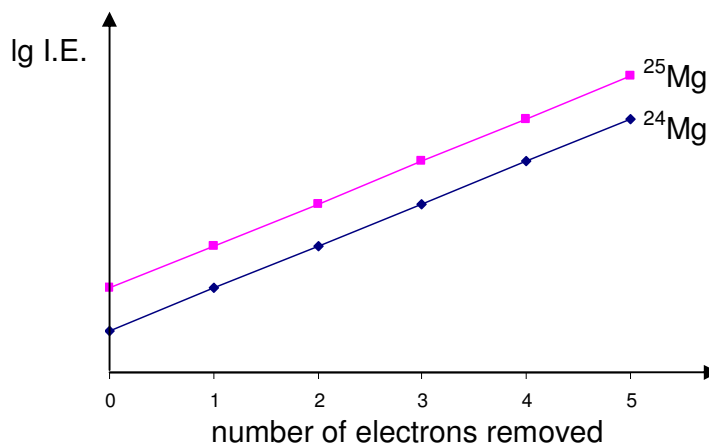
(ii) Magnesium exists as three isotopes.

Mass number	Relative abundance (%)
24	79
25	10
y	11

Given that the relative atomic mass of magnesium is 24.32, determine the mass number **y**.

[3]

(b) A student was asked to sketch the first five ionisation energies of magnesium for ^{24}Mg and ^{25}Mg . Without referring to the *Data Booklet*, the student produced the graph as shown.



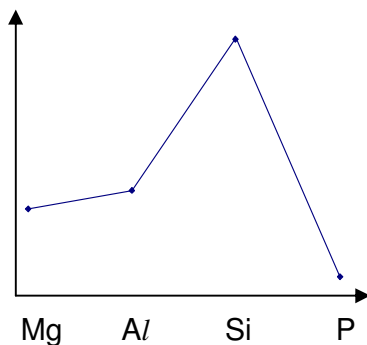
Identify and explain **two** mistakes made in the graph drawn.

.....

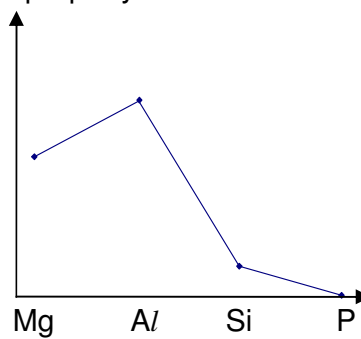
[2]

- c) State and explain the physical properties of the following graphs for the elements Mg to P.

Physical property **A**



Physical property **B**



Physical property **A**:

.....

Physical property **B**:

.....

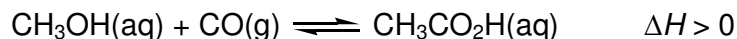
[5]

[Total: 10]

[Turn over

2 Ethanoic acid, an important industrial commodity chemical in the world, finds a variety of uses in vinegar, polymer formation and ester formation.

(a) The preferred industrial method for the manufacture of ethanoic acid is by carbonylation of methanol which accounts for 60% of the total world ethanoic acid manufacturing capacity.



The process uses a rhodium-based catalyst and operates at a temperature of 150–200°C and a pressure of 30–60 atm, giving a selectivity of greater than 99%.

(i) Using Le Chatelier's Principle, explain why the above high pressure is chosen as the process condition.

.....

.....

.....

.....

(ii) The above process was carried out in an experiment in a 2 dm³ reaction vessel. When equilibrium was reached at a given temperature, the mixture contains 3.66 mol of methanol, 4.68 mol of carbon monoxide and 2.66 mol of ethanoic acid. Calculate K_c at this temperature.

(b) Ethanoic acid is a weak acid. In one experiment, 25.0 cm^3 of $0.200 \text{ mol dm}^{-3}$ NaOH is titrated against ethanoic acid of the same concentration until equivalence point is reached.

(i) Write an equation for the reaction between ethanoic acid and NaOH.

.....

(ii) Predict the pH value at the equivalence point. Explain your answer.

.....

.....

.....

.....

(iii) When a certain volume of ethanoic acid is added to the mixture after the equivalence point has been reached, the solution is found to be able to resist pH changes when a small amount of acid or alkali is added. Explain why this is so. Illustrate your answers with equations.

.....

.....

.....

.....

.....

.....

.....

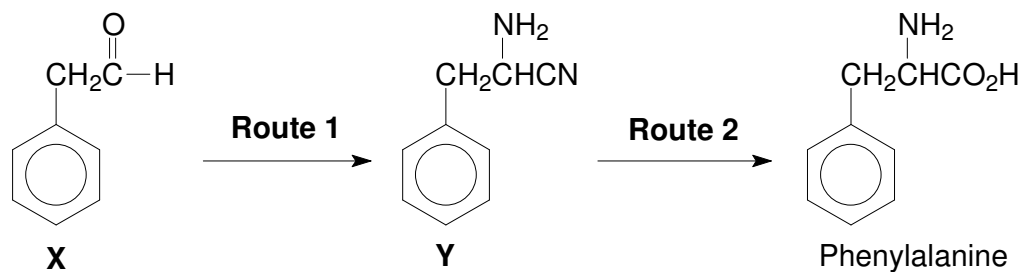
.....

[6]

[Total: 10]

[Turn over

- 3 The following represents part of the synthetic route of the amino acid, phenylalanine.



- (a) Compound **W**, a structural isomer of **X**, gives a yellow precipitate on warming with alkaline aqueous iodine. Suggest a possible structural formula for **W**.

[1]

- (b) The formation of **Y** from **X** in **Route 1** involves **three** steps. Suggest the reagents and conditions for each step and the structures of the intermediates formed.

[5]

- (c) State the type of reaction occurring in **Route 2**.

.....[1]

- (d) Only **one** of the compounds, **X**, **Y** or phenylalanine will react with **each** of the following reagents. In **each** case, identify the compound concerned and draw the structural formula of the organic product formed.

[Assume no reaction with the -NH_2 group.]

reagent	compound X , Y or phenylalanine	structural formula of organic product
$\text{CH}_3\text{CH}_2\text{OH}$, concentrated H_2SO_4		
Tollens' reagent		

[3]

[Total: 10]

[Turn over

- 4 Nitrogen compounds have many industrial uses. One of the most common compounds, ammonia, is produced in the Haber Process and fertilizer generated from ammonia sustains one third of the world's population.

A lesser known process, the Ostwald Process, is used in the manufacture of nitric acid and takes place in three stages.

- I Ammonia is mixed with air and passed over a catalyst at 900°C to give nitrogen monoxide and steam.
- II Nitrogen monoxide is then cooled and mixed with air to give a brown gas, nitrogen dioxide.
- III Nitrogen dioxide is then passed through a water spray in a steel absorption tower to yield nitric acid and nitrogen monoxide.

80% of nitric acid manufactured by the Ostwald Process is used to produce ammonium nitrate, which is used as fertilizers. Ammonium nitrate is formed by the reaction of nitric acid with aqueous ammonia. Decomposition of ammonium nitrate produces nitrous oxide, N_2O , and steam.

A company that manufactures Brand X fertilizer, which contains a mixture of ammonium nitrate and other chemical compounds, claims that their fertilizer contains the highest concentration (65% by mass) of ammonium nitrate available on the market today. A farmer decided to test if this claim is indeed true. He heated a 450 g sample of this fertilizer and found that 80 dm^3 of nitrous oxide was released at room temperature and pressure.

- (a)(i) Write balanced equations for the reactions occurring in stages I and III.

.....
.....

- (ii) What are the changes in the oxidation states of nitrogen in stage III of the Ostwald Process?

.....
.....

[3]

(b)(i) Determine if the company's claim about Brand X fertilizer is valid.

(ii) Hence calculate the mass, in kg, of nitric acid required to manufacture 10 kg of Brand X fertilizer.

[5]

(c) With the aid of relevant half-equations, write the balanced equation for the reaction between aluminium and concentrated nitric acid. The reaction produces NH_4^+ as one of the products.

.....

.....

.....[2]

[Total: 10]

[Turn over

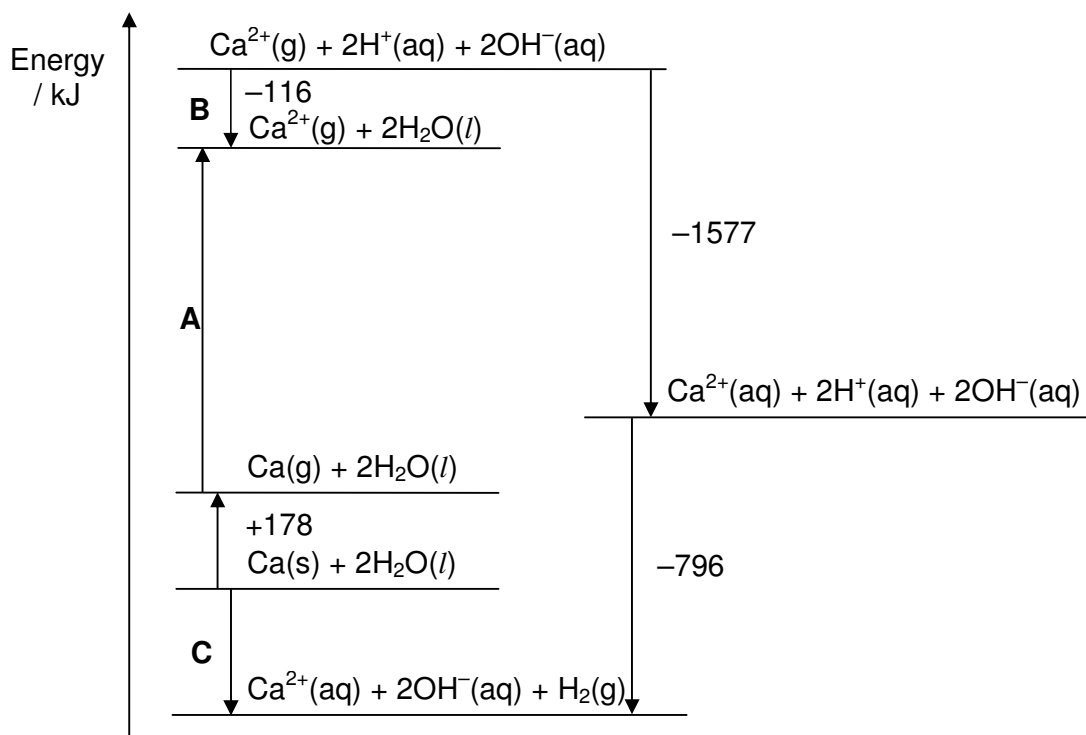
Section B

Answer any **two** questions. Begin each question on a fresh sheet of paper.

5

(a) State Hess' Law. [1]

(b) Calcium reacts with water to form aqueous calcium hydroxide. The energy cycle for the formation of calcium hydroxide, Ca(OH)_2 , is shown below.



- (i) Name the enthalpy changes involved in steps **A** and **B**.
- (ii) Use the *Data Booklet* to obtain the value of the enthalpy change for step **A**.
- (iii) Calculate the enthalpy change for step **C**, i.e. for the reaction

$$\text{Ca(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Ca}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) + \text{H}_2(\text{g})$$
- (iv) In an experiment, 0.50 g of calcium is added to 200 cm³ of water. The temperature rise is recorded. Using your answer to (iii), calculate the expected temperature rise for the reaction.
- [Assume the heat capacity of all solutions = 4.2 J K⁻¹ cm⁻³]
- (v) Write an equation for the lattice energy of solid calcium hydroxide.
- (vi) Comment on the relative numerical value of the lattice energy of calcium hydroxide, Ca(OH)_2 and barium hydroxide, Ba(OH)_2 .

[9]

- (c) Both calcium and beryllium are Group II elements. However, their compounds differ greatly in their boiling points.

Compound	Boiling point / °C
BeCl_2	482
CaCl_2	1940

- (i) Explain the difference in their boiling points.
- (ii) Draw dot-and-cross diagrams to illustrate the bonding in these two compounds.

[5]

- (d)(i) Calcium chloride has a similar formula mass to Cl_2O . State and explain two differences in the physical properties of Cl_2O as compared to CaCl_2 .

- (ii) Draw a diagram to illustrate the shape of Cl_2O . State the shape and indicate the bond angle.

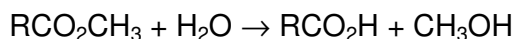
[5]

[Total: 20]

[Turn over]

6

- (a) The kinetics of the hydrolysis of the ester, RCO_2CH_3 , may be investigated by determining the concentration of RCO_2H produced.



In a 1 dm^3 mixture, 0.350 mol of the ester was hydrolysed by heating with water and using hydrochloric acid as catalyst. The following results were obtained.

Time / s	Concentration of RCO_2H / mol dm^{-3}
0	0
340	0.105
680	0.185
1080	0.243
1440	0.278

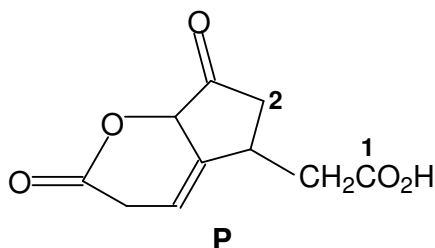
- (i) By drawing a suitable graph using the data given above, show that the reaction is first order with respect to the ester.

It has been found that the hydrolysis reaction is first order with respect to the hydrochloric acid.

- (ii) Write down the rate equation for this reaction and state the units of the rate constant.
- (iii) State how the half-life of the reaction is affected when the experiment is repeated with the following changes.
- (I) Concentration of HCl is doubled.
- (II) Concentration of RCO_2CH_3 is doubled.
- (iv) With the aid of a sketch of the Boltzmann distribution, explain how the presence of a catalyst increases the rate of a chemical reaction.

[11]

- (b) Compound **P** contains a few functional groups, one of which is the ester functional group.



- (i) Account for the acidity of **P**.
- (ii) The geometry (shape) with respect to carbon atoms labeled **1** and **2** are different. State the shape and type of hybrid orbitals at carbon atoms labeled **1** and **2**.

Hence sketch the shapes of the hybrid orbitals around the two carbon atoms.

- (iii) Suggest the structural formulae of the products formed when **P** reacts with

(I) H_2 over Pt catalyst,

(II) hot aqueous NaOH.

[9]

[Total: 20]

[Turn over

7

- (a) The oxides of sodium, silicon and phosphorus are different in their bonding. Describe the reaction of these oxides with water, relating them to their structure and bonding. Write equations where appropriate. [4]
- (b) Sodium metal is commonly used in organic reactions such as those between sodium and alcohols.
- (i) Write an equation to show the reaction between sodium and butan-1-ol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$.
- (ii) Draw the structures of the other three alcohols which are isomeric with $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$.
- (iii) Hence, suggest how the alcohols in (ii) can be distinguished from one another by oxidation, followed by subsequent testing of the oxidation products. [5]
- (c) An alcohol **U**, $\text{C}_{10}\text{H}_{14}\text{O}$ is oxidised by hot acidified potassium dichromate to give **V**, $\text{C}_{10}\text{H}_{12}\text{O}$, which gives an orange precipitate with 2,4-dinitrophenylhydrazine but no silver mirror with Tollens' reagent.

However, **U** is oxidised by hot acidified potassium manganate(VII) to give a different compound **W**, $\text{C}_7\text{H}_6\text{O}_2$, which liberates carbon dioxide when sodium hydrogencarbonate is added.

When **U** is boiled with an excess of concentrated sulphuric acid, **X** is produced. Treatment of **X** with hot acidified potassium manganate(VII) gave only two products, **W** and **Y**. **Y**, $\text{C}_3\text{H}_6\text{O}$, gave a pale yellow precipitate on warming with alkaline aqueous iodine.

Compound **U** can be made by heating **Z**, $\text{C}_{10}\text{H}_{13}\text{Br}$ with aqueous sodium hydroxide.

- (i) Deduce the structures for each lettered compound, **U** to **Z**, and give an account of the chemistry involved.
- (ii) Write a balanced equation for the reaction between **V** and 2,4-dinitrophenylhydrazine.

[11]

[Total: 20]